LOANFINDER

Modern Loan Comparison & P2P Lending Platform

PROJECT STATUS REPORT

A complete overview of what has been built, how it works,
and where the project currently stands.

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HackIndia 2026 Demo Team

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1. Executive Summary

LoanFinder is a browser-based fintech web application built as a self-contained demo for HackIndia 2026. It lets individual borrowers in India compare advertised loan rates side-by-side across 5 real Indian banks (7 published offers), calculate exact EMIs, check eligibility, locate branches on a live map, follow a document checklist, and additionally borrow or lend money in a built-in peer-to-peer (P2P) marketplace. The working pair "consumer borrower + P2P lender" is the core concept the team presents to judges.

The entire application runs as a static vanilla HTML/CSS/JavaScript codebase in the folder p1/ — no framework, no build step, no server required. It is fully offline-capable (Lucide icons and the Leaflet map library are vendored locally) and works by simply opening index.html or serving the folder with any static server. All app state is held in memory and persisted to the browser's localStorage.

Starting rate	Partner banks	Status
7.85% p.a.*	5	Working demo

The catalogue contains 5 real Indian banks (State Bank, HDFC Bank, ICICI Bank, Axis Bank, Kotak Mahindra) publishing 7 verified offers across 2 loan categories. Rates shown are advertised starting rates, not quotes. Any scale or security claim such as "25+ banks", "25,000+ borrowers" or "₹1,500 Cr+ facilitated" would be unsupported and is deliberately not made.

2. Product Purpose & Positioning

Problem being solved

Choosing a loan in India means visiting many bank websites, comparing confusing advertised rates, hidden charges and document requirements, and having no single place to track it all. LoanFinder turns that multi-day research into a single dashboard where a visitor goes from "which loan should I pick" to a confident, comparison-backed loan decision (or a completed P2P funding match) in minutes.

Core users

- Consumer borrowers comparing home, personal and auto loans across banks to find the best advertised rate and lowest EMI before applying.
- P2P marketplace participants — borrowers posting funding requests and lenders funding vetted borrowers.

What makes it different

The combination a neighboring product could not truthfully copy: side-by-side comparison of advertised bank offers (not invented aggregate rates) joined with branch-level discovery, an eligibility checker, document readiness, hidden-charge transparency, and a built-in P2P lending marketplace — all in one interface.

3. Technology & Architecture

Aspect	Detail
Platform	Web (browser), no-build static demo — runs offline except the branch map basemap, which needs internet
Stack	Vanilla HTML5 + CSS3 + JavaScript (ES6) — no framework, no build step
Files	p1/index.html · p1/styles.css · p1/app.js · p1/README.md
Vendored libraries	Lucide icons (safeIcons wrapper), MapLibre GL (primary map

	engine) and Leaflet (fallback) — saved in p1/vendor/ so the app loads with no CDN. Basemap tiles are still fetched at runtime from openfreemap.org and Esri
State management	Single appState object in memory plus a tiny localStorage document store ("LoanFinder DB v1")
Fonts / style	Plus Jakarta Sans preference; primary blue #2563eb, navy sidebar #0f172a, emerald #10b981
Number conventions	Indian ₹ formatting with lakh/crore; English copy
Run method	Double-click index.html OR any static server (python -m http.server 3000 or npx serve .)

EMI formula used throughout: EMI = P · r · (1+r)^n / ((1+r)^n − 1), where r is the monthly rate and n the number of months. The same engine powers the comparison cards, eligibility checker and P2P estimated-return previews.

4. How the App Works (User Flows)

Borrower / comparison flow

1. Land on the public landing page Sign In (demo auth) enters the dashboard.
2. Choose loan type(s) — Home, Personal, Car, Education, Business or Gold.
3. Set loan amount (slider ₹1 lakh to ₹1.5 crore), tenure, annual income, employment type and credit score.
4. Press "Find Matching Loan" — the top-3 ranked recommendations unlock (scored by rate, credit-fit, income-fit and fee).
5. Open any bank's details page: Overview, Documents checklist (with DigiLocker import), Eligibility guidelines, Hidden Charges with a full cost summary.
6. Locate branches on the Google-Maps-style map (smooth zoom, rotate, 3D pitch, Map/Satellite switcher), filtered by bank / city / search, with directions and call actions. Distances are shown only from your real GPS position once you grant permission; no live open/closed status is claimed.
7. Bookmark offers into "Saved" for later comparison.

P2P flow

1. Enter "P2P Lending" and choose between Borrow Money or Lend Money.
2. Borrower: fill the loan request form (amount, purpose, rate, income, credit) post it to the marketplace, see ranked lender matches, and chat with interested lenders.
3. Lender: set preferences (amount, max rate, risk tolerance, tenure) ranked borrower matches, full funding preview with estimated returns, confirm a demo investment.
4. Track the portfolio (invested, expected returns, interest earned), view repayment schedules, and chat with borrowers.

5. Features Implemented

#	Feature / Module	What it does
1	**Landing page**	Hero value prop, honest trust stats (7.85% starting, 5 banks, 7 live offers, 5 cities), features grid, "How it works" steps, CTA
2	**Demo Auth (Login / Sign Up)**	Modal and full-page login/signup with password toggle, remember-me, "safe &

		secure" badges
3	**Sidebar navigation**	Dark navy sidebar, active indicator, notification badge, P2P open-request counter, logout
4	**Multi-loan type selection**	Selectable cards for Home, Personal, Car, Education, Business, Gold with starting rates
5	**Loan requirements panel**	Amount slider (real-time display), tenure dropdown, annual income, employment type, credit score
6	**Comparison engine**	Sorts offers by lowest advertised rate, EMI computed live; global search across banks/rates/tags
7	**Top-3 Recommendation ranking**	Scoring: rate 40%, credit-fit 25%, income-fit 20%, processing fee 15%. Shows % match, notes, EMI
8	**Best-offer pinned card**	Emerald "Best Match: Lowest Rate" highlight above the full comparison
9	**Full comparison + sorting**	Sort by rate, EMI, processing fee, or max amount; empty states handled
10	**Bank details & tabs**	Overview, Documents, Eligibility criteria, Hidden Charges for each bank offer
11	**Document checklist**	Categorized list (Identity, Address, Income, Tax, Property, Photos) with Ready / In Review / Required statuses
12	**DigiLocker integration (demo)**	Aadhaar → OTP → fetch flow, select & import verified documents, marks checklist items verified
13	**Hidden charges transparency**	Per-offer fees (legal, stamp duty, bounce, prepayment, penal interest) + "Your Real Loan Cost" summary
14	**Branch locator**	Google-Maps-style MapLibre map (Map/Satellite switch, fit-all, InfoWindow cards) + side cards, bank/city filters, GPS distance, phone, Google directions. No live open/closed status
15	**Eligibility checker**	Income/EMI/credit/employment sliders, max eligible amount via FOIR rule, per-loan breakdown, personalized tips
16	**Saved offers**	Bookmark bank offers and P2P requests; persists across reloads in localStorage
17	**Profile & settings**	Pre-filled applicant profile; notification toggles; one-click "Reset Demo Data"
18	**Support chatbot widget**	In-app floating helper with quick replies about loans, EMI, eligibility, P2P, branches

19	**P2P Lending marketplace**	Borrow/lend modes, marketplace feed, funding progress bars, risk badges, match scores
20	**P2P Portfolio & chat**	Investment tracking with amortization schedules, expected vs earned interest, per-loan chat with peers

6. The P2P Lending Marketplace

The P2P side is a full two-sided demo: a borrower posts a request, lenders discover and fund it, and both sides track their position in a portfolio with a chat thread. It carries equal design weight to the comparison engine, as documented in the product's brand commitments.

Area	Status
Marketplace feed	Seeded borrower requests (medical, wedding, business, education, renovation) with purposes, risk badges and match scores
Borrow money	Request form (amount ₹5k–₹5L, rate 5–24%, tenure, income, credit) → post → ranked lender matches → chat
Lend money	Lender preferences → ranked matches → lend modal with live estimated-returns preview → confirm demo investment
Portfolio	Summaries (total invested, expected returns, interest earned, active loans) + per-investment amortization and chat
Chat	Borrower ↔ lender, borrower ↔ investor, and user ↔ recommended lender; persisted messages
Persistence	Stored in localStorage under loanfinder.p2p.v1 and the shared LoanFinder DB (reset as one unit)
Validation rules	Minimum lending ₹1,000, can't exceed remaining funding, credit-score based risk tiering, confirm-before-delete for requests

7. Data Model & Persistence

Everything ships as static data inside app.js — no external database is required for the demo.

Dataset	Contents
LOAN_TYPES	6 loan categories with starting rates and popularity flags
BANK_OFFERS	Advertised offers for State Bank, HDFC, ICICI, Axis, Kotak (home + personal) with rates, fees, max amount/tenure, eligibility, employment rules
HIDDEN_CHARGES	Per-offer fine-print items with amounts and % fees, plus informational rows
DOCUMENTS_DATA	Checklist categories with icons, descriptions and status
DIGILOCKER_DOCS	Sample verified documents (Aadhaar, PAN, Driving License, etc.)

	for the import demo
BRANCH_DATA	Geo-coded branches across Mumbai, Delhi & Bengaluru with hours, phone, open status
P2P data	Demo borrower requests, lender profiles, investments and chat threads
localStorage DB	loanfinder.db.v1 (savedOffers, savedP2P, p2p, meta) — read/written via a tiny db helper

Note: README.md documents where the static BANK_OFFERS and BRANCH_DATA blocks would be swapped for live backend endpoints (/api/loans/compare, /api/branches, /api/documents/upload) in future work.

8. Demo Data & Constraints

- Rates and EMI outputs are computed from the published advertised rates. The bank count shown anywhere in the product (5 banks, 7 offers) matches the actual catalogue exactly; inflated counts such as "25+ banks" are never displayed.
- Real Indian bank names and their offer cards are trusted content kept exactly as-is.
- Unsupported trust claims ("Bank-Grade Security", "256-bit bank-level SSL", "25,000+ borrowers") were deliberately removed. There is no server, so an SSL claim could not be true here. The card now reads "Your Data Stays Local", which is accurate for a localStorage demo.
- Branch records for Hyderabad and Pune are attributed to official bank locators with IFSC codes and source links; coordinates are only plotted where an OpenStreetMap node of the same bank was found in the same locality. Records without verified coordinates are listed but not plotted. Older Mumbai / Delhi / Bengaluru records are clearly labelled illustrative demo data.
- The app is static and self-contained; it must keep working by opening index.html.
- No real customer testimonials, case studies, press mentions or audited statistics exist today — future work must not manufacture additional formal evidence.

9. Current Stand & Verification

Where the project stands right now

LoanFinder is a complete, working, self-contained demo. All listed modules render and function in a modern browser from a single static folder. The comparison engine, eligibility logic, branch map, document checklist and the full two-sided P2P marketplace (post match fund track chat) are implemented and run end-to-end.

Verification checklist

Item	Check	Status
1	App opens directly from index.html (no build step; only the map basemap needs internet)	Done
2	Loan comparison, EMI and top-3 ranking work live	Done
3	Bank details: docs, eligibility, hidden charges	Done
4	Branch locator map with filters & directions	Done
5	Eligibility checker with live recomputation	Done
6	Saved offers persist after reload	Done
7	P2P: post request, match, fund, portfolio, chat	Done
8	Demo data reset restores the original seed	Done

10. How to Run

1. Open the folder p1/ in the repository.
2. Double-click index.html (or right-click → Open with Chrome / Edge).
3. Recommended dev server: python -m http.server 3000 then open http://localhost:3000 — or npx serve .
4. Suggested demo path for judges: landing → Sign In → Home loan → Find Matching Loan → SBI details & Docs → Hidden Charges → Branches map → Saved → P2P Lending (Lend Money → fund a borrower) → Portfolio → Chat.

Suggestions: use the Reset Demo Data button in Settings before each new demo so judges always start from the clean seeded state.

11. Next Steps

Priority	Planned improvement	Type
High	Real backend endpoints feeding BANK_OFFERS / BRANCH_DATA (API stubs are documented in README.md)	Backend
High	Record a polished walkthrough / demo script to rehearse the judge's path	Demo prep
Medium	Connect DigiLocker import to the real DigiLocker APIs	Integration
Medium	Add remaining loan categories (car, education, business, gold) to BANK_OFFERS for full coverage	Content
Medium	Formal evidence: testimonials, case studies or audited stats — only if available and truthful	Validation
Low	Unit tests / linting for app.js to harden the demo build	Engineering
Low	Deploy the static folder to any host for a shareable live URL	Deployment

Prepared automatically from the project source (index.html, app.js, styles.css, README.md, PRODUCT.md).